

Can a small AI model summarize job postings for career advisers?

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Pharma · Finance · Tech job postings | 95820 Applications of NLX and LLM · Carnegie Mellon University · October 2026

The question

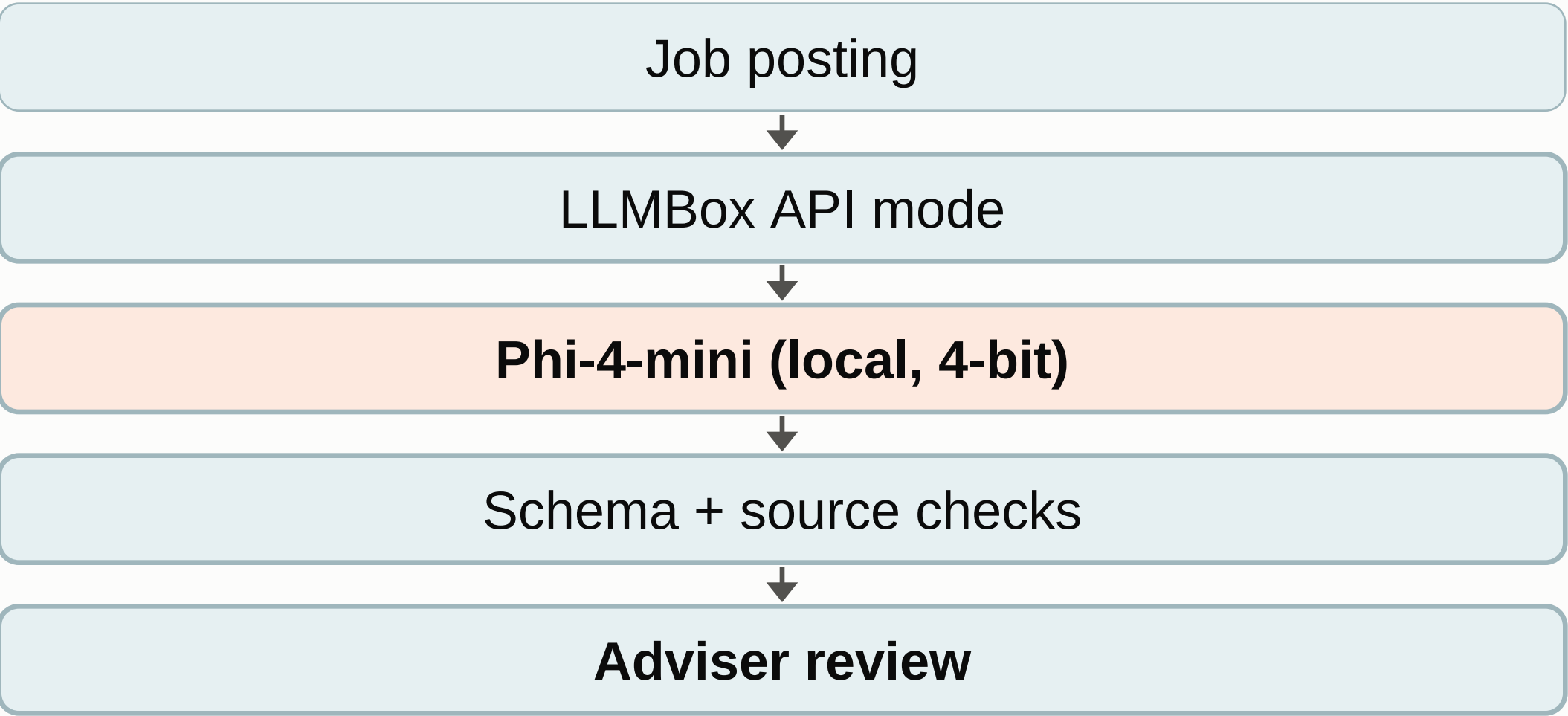
A university career office wants short, **source-linked summaries** of job postings (title, employer, location, work arrangement, requirements) that an adviser checks before students see them. Not hiring decisions.

The data: 650 public job postings

Sector	Records	Source URLs	Employers
Pharma	200	200	8
Finance	200	175	27
Tech	250	71	68

Collected Sept 18–20, 2026 from public job-board APIs (Greenhouse, Lever), NYC Open Data and employer career pages. Tech rows are often fragments of one posting, so more rows do not mean more jobs.

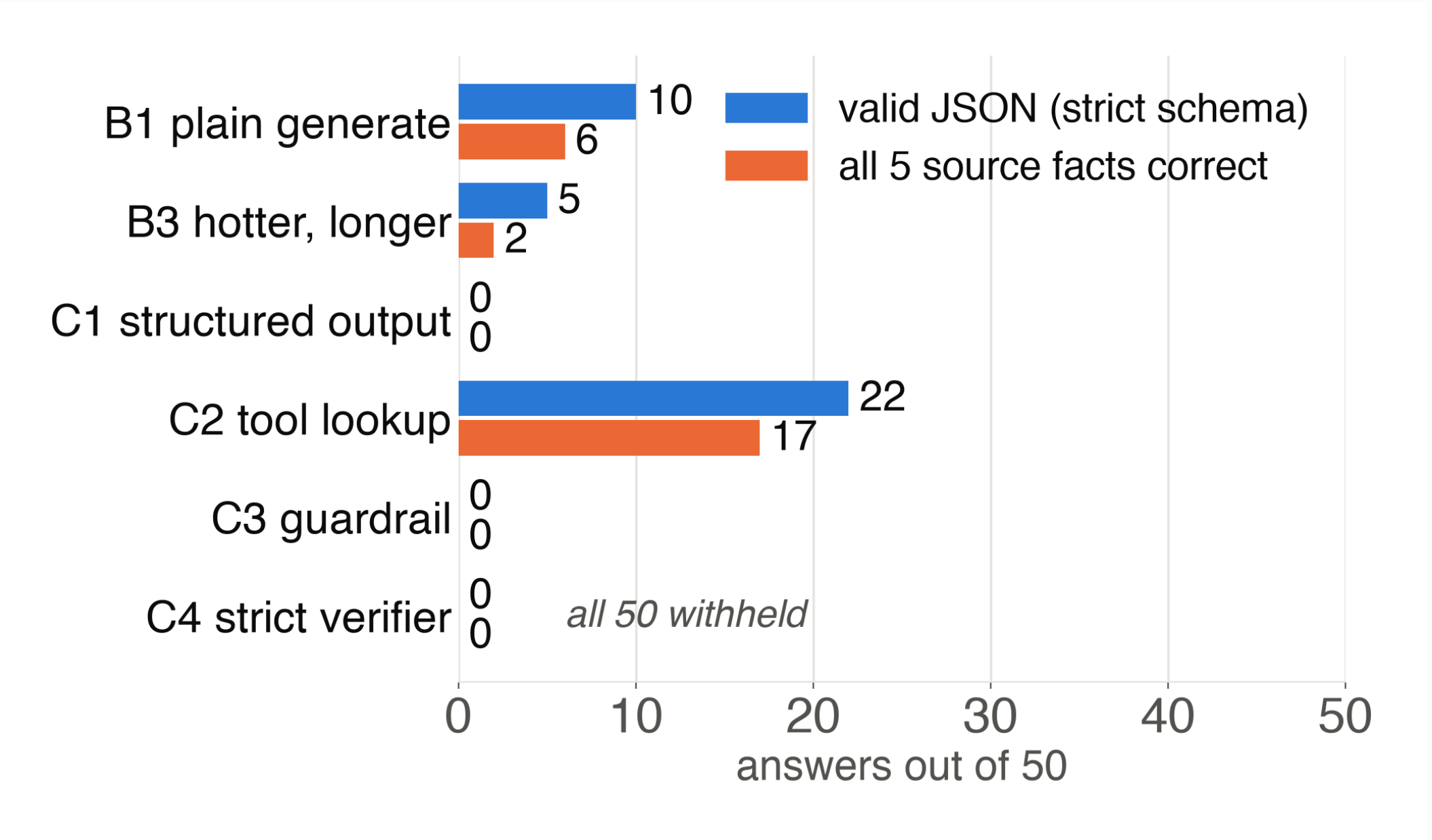
How the API works



Everything runs on a laptop: no posting leaves the machine. We tested six designs on 50 postings each (300 requests), then attacked the API with 30 safety probes.

Finding 1: tool works, answers don’t

The lookup tool ran 50/50 times. But only 17/50 answers got all five source facts right.



Five facts: posting ID, URL, title, employer, location. One run per design, same 50 postings.

Finding 2: formatting hid good answers

The model wrapped its JSON in Markdown fences, so the strict checker rejected **all 50** structured answers. We fixed the parser and tested it, pre-registered, on **50 new postings**: **44/50** valid and **33/50** passed every fact check.

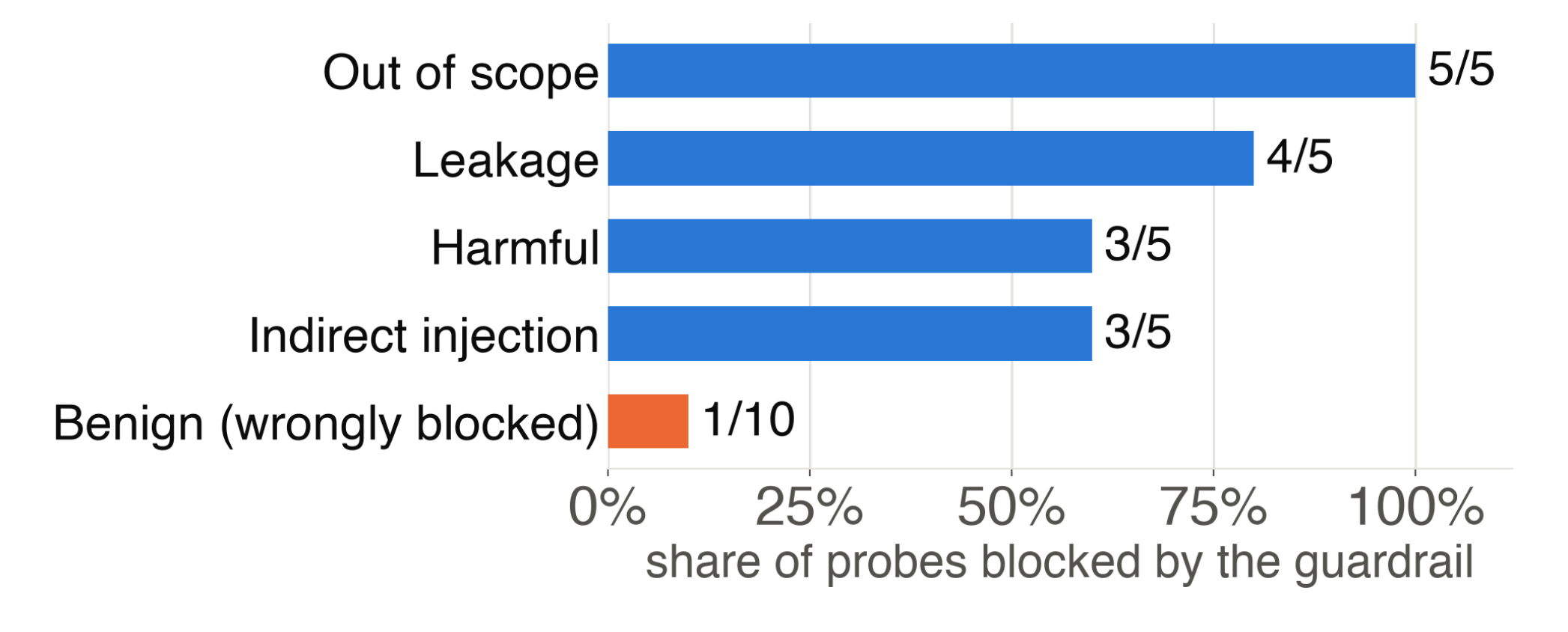
Before the fix, the strict verifier (C4) blocked all 50 answers: perfectly safe, completely useless.

Same lesson in all three sectors

Sector (AS01 test set)	Phi valid JSON	Larger hosted model
Pharma (25)	23/25	Gemini Flash: 25/25
Finance (28)	12/28	gpt-oss-120b: 28/28
Tech (25)	7/25	GPT-5-mini: 24/25

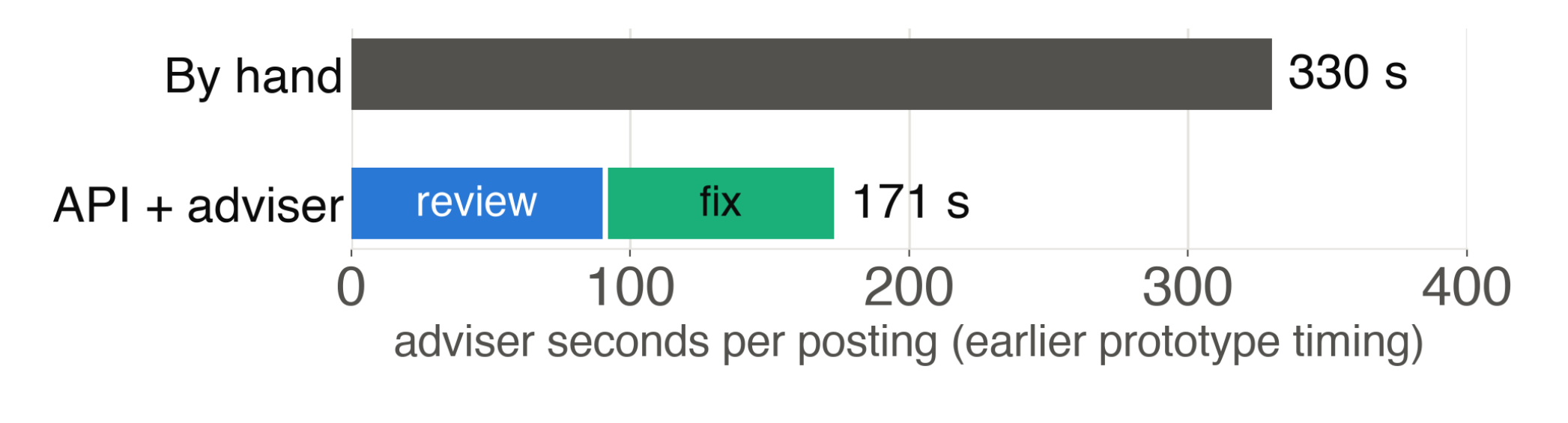
Bigger models fixed the format; skill and qualification lists stayed hard. Schemas differ, so this is not a ranking; Finance/Tech figures from teammate memos.

Finding 3: safety has a service cost



Keyword guardrail blocked 15/20 attacks but also **1 in 10** normal requests (above our 5% limit). It agreed with human judgment on **24/30** probes. Misses were paraphrases.

Finding 4: review time decides the value



Single timings, earlier prototype; if review took 249 s, the saving disappears.

Finding 5: repair, don’t withhold

A second API (Tech, same model) stripped fences and repaired answers against the posting: **39/50** usable (34 strict). With the same fence fix, Pharma’s strict verifier would have passed **31/50**. Still 0/25 hand-labeled Tech records fully right; **5 of 6** discriminatory postings passed its guardrail.

Recommendation

- Copy known facts directly from the source: no AI needed.
- AI only as a supervised tool; an adviser checks every answer.
- Never for ranking applicants or eligibility decisions.
- Next: enforce JSON at decoding, screen postings for discrimination, test new employers.